
Dual Receptive-Field Routing: Robust Missing-Evidence Vision with Fixed-Capacity Depth

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Abstract

Feedback to visual cortex can generate a receptive field outside a neuron’s feed-forward center, supplying contextual evidence where direct sensory drive is absent. We translate this narrow computational observation into Dual Receptive-Field Routing (Dual-RF), a parameter-neutral intervention for vision models with explicit missingness. A deterministic, center-excluded surround fills missing locations only where observed context exists, while clamping every observed value exactly; an adaptive variant expands through a fixed field ladder when local support is absent. The completed image and observation mask enter an official pretrained Swin V2-Tiny. All examples execute through an early head; a fixed-capacity top- k subbatch executes the final Swin stage as one dense gather–compute–scatter operation. Baseline and Dual-RF contain exactly 27,596,414 learned parameters and share the checkpoint, data order, masks, initialization, and optimization schedule.

We run paired three-seed studies on natural-color CIFAR-10 and MNIST. Each uses 6,000 training and 4,000 held-out examples; training masks remove 15–25% of rows, blocks, or pixels, while evaluation masks are more severe. On CIFAR-10, a 25% final-stage route changes clean, 45%-row, 35%-block, and 45%-pixel accuracy by +2.275, +12.325, +1.542, and +27.042 percentage points. End-to-end Apple M4 MPS latency is 29.2% lower. The row, block, and pixel paired intervals exclude zero, although $n = 3$ remains low power. MNIST changes by +0.517, +4.825, -0.417, and +4.400 points and is 30.7% faster, retaining a negative block boundary. A parameter-identical single- 3×3 surround recovers nearly all CIFAR gain and is 31.5% faster; adaptive field expansion adds only +2.217 mean points on rows and is not significant. The supported contribution is therefore evidence-preserving surround completion plus executable fixed capacity, not a universal advantage for multi-scale reach, biological equivalence, broad state of the art, or measured energy reduction.

1 Introduction

Modern visual backbones apply nearly the same sequence of dense operations to every example. This uniform schedule is convenient for accelerators, but it conflates three questions: what sensory evidence is available, what contextual evidence can replace a missing input, and how much depth a particular example requires. Biological vision separates these questions across pathways and time. Feedforward drive establishes a classical receptive field; recurrent and feedback pathways add context outside that field; and downstream activity depends on the interaction of those sources rather than a uniform replay of every circuit (Keller et al., 2020; Fişek et al., 2023). These observations do not provide a literal neural-network blueprint, but they do suggest an engineering decomposition that can be falsified.

This report studies that decomposition under explicit missing visual evidence. Let x denote an image and m an observation mask. A useful architecture should preserve $m \odot x$ exactly, obtain a contextual estimate only where $m = 0$, and expose an expensive path whose execution can be measured. We implement those requirements as Dual Receptive-Field Routing (Dual-RF, ours). Its first component is a nonparametric center-excluded surround. The core local operator fills only missing pixels with observed 3×3 support; an adaptive-reach variant selects the smallest supported field from $(3, 5, 9, 17, 31)$. Its second component is an early/deep Swin cascade: all examples execute stages 1–3 and an early head, while a declared fraction ρ executes the final stage in one contiguous subbatch.

The distinction between representation and routing is central. A dense Dual-RF model tests whether complementary fields improve robustness. A capacity sweep tests whether the final stage can be skipped in an executable way. Entropy, random, and label-using oracle controls test whether routing intelligence matters. A parameter-identical single- 3×3 field tests whether adaptive reach matters. These comparisons keep learned parameters, checkpoint, heads, optimizer, data, and masks identical.

The primary result uses an official pretrained Swin V2-Tiny (Liu et al., 2022b). Across three paired CIFAR-10 seeds, the 25% route improves row and pixel missingness by +12.325 and +27.042 points, improves blocks by +1.542 points, and reduces measured latency by 29.2%. MNIST independently improves rows and pixels but not blocks. A smaller ConvNeXt-style replication yields modest accuracy effects and a 15.6% latency reduction. The direct single-field control changes the interpretation: most absolute robustness comes from local evidence-preserving completion, while broad adaptive reach has no general advantage in this protocol.

Our contributions are:

1. We formulate **complementary supported fields**, a deterministic center-excluded completion operator motivated by measured feedback receptive fields but explicitly separated from claims of biological equivalence.
2. We integrate the operator into a **strong pretrained Swin V2-Tiny** with exact learned-parameter matching and a fixed-capacity final-stage route that executes as one dense subbatch.
3. We report a **cross-dataset paired audit**: CIFAR-10 and MNIST, three seeds, three severe mask structures, a capacity sweep, routing controls, and end-to-end device latency.
4. We isolate novelty with a **parameter-identical single-field control**. It shows that local completion explains most gains and that adaptive reach is, at best, row-specific in the current design.
5. We retain negative evidence: the MNIST block mean is negative, adaptive reach is not generally superior, entropy is close to random, and no broad state-of-the-art or energy claim is made.

2 From neuroscience measurement to engineering abstraction

2.1 Complementary receptive fields

Keller, Roth, and Scanziani (Keller et al., 2020) isolated cortical feedback while recording neurons in mouse visual cortex. They found that feedback can generate a second receptive field outside the feedforward receptive field. The two fields are spatially offset and functionally complementary: the feedback field can report differences between spatial regions and provide an estimate when feedforward evidence is absent. This is more specific than the general statement that “brains predict.” It suggests a center/context decomposition with an observable condition under which context is recruited.

Fişek et al. (Fişek et al., 2023) measured feedback-related dendritic events and showed that retinotopic alignment changes feedback influence. Aligned feedback tended to be suppressive at the receptive-field center, while displaced feedback could facilitate apical dendritic activity. Dual-RF borrows only the separation of center and surround. It does not model apical voltage, spikes, calcium dynamics, cortical layers, or the sign of synaptic feedback.

The operator is therefore an abstraction with three invariants:

1. **Center exclusion**: a location cannot predict itself.

2. **Evidence preservation:** observed values are copied exactly.
3. **Conditional reach:** a broad field is used only when a smaller field has no observed support.

These invariants can be tested independently of task accuracy. They are weaker and more auditable than claiming a cortical simulation.

2.2 What the imaging on the project page means

The project page contains two real biological visualizations. One is a preprocessed human fMRI BOLD run from OpenNeuro ds000228 (Richardson et al., 2018); the second is a Creative Commons excerpt of in-vivo cortical calcium imaging from the CaImAn study (Giovannucci et al., 2019). Neither dataset trains, calibrates, or evaluates Dual-RF. BOLD voxels are slow hemodynamic measurements, whereas calcium fluorescence is a cellular activity proxy. They are presented to distinguish scales and provenance, not to imply that model states were decoded from either modality. The architecture diagram is rendered separately for the same reason.

3 Related work

Predictive and feedback computation. Predictive coding models explain extra-classical receptive-field effects through top-down prediction and bottom-up residuals (Rao and Ballard, 1999). Recurrent vision systems have also been used to recognize occluded objects and to explain away competing hypotheses. Dual-RF is narrower: it assumes an explicit observation mask and uses a fixed, non-learned contextual estimator. It does not perform iterative generative inference.

Masked completion. Partial convolution renormalizes convolutional responses by the number of valid pixels and updates a validity mask during image inpainting (Liu et al., 2018). Normalized convolution and diffusion-based filling are older, closely related ideas. Dual-RF does not claim that masked normalization is new. Its distinct operator chooses the *smallest supported center-excluded field* from a fixed ladder and clamps sensory evidence, then tests that representation inside a capacity-routed pretrained classifier. A single-scale field is a direct ablation, not a straw baseline.

Conditional depth. Adaptive Computation Time allocates recurrent steps by input difficulty (Graves, 2016); Skip RNN learns to skip state updates (Campos et al., 2018). DynamicViT prunes tokens to reduce vision-transformer FLOPs and reports throughput gains (Rao et al., 2021). Mixture-of-Depths (MoD) uses fixed-capacity top- k routing so tensor shapes and total compute remain predictable (Raposo et al., 2024). The fixed-capacity idea in Dual-RF is directly indebted to MoD. Our route is over examples at one late visual stage rather than over tokens at many language-model layers.

Surprise-based fixed-capacity routing is also not unique to this project. Subjective Depth and Timescale Transformers route using Bayesian surprise and a static top- k budget (Wieser et al., 2025). That close prior invalidates any claim that a generic prediction residual plus fixed top- k is novel. Our audit consequently demotes prediction mismatch after it fails to beat a confidence-only router. The contribution is the complementary-field representation and its paired hardware integration, not a new general theory of surprise routing.

Backbones. ConvNeXt modernized convolutional networks using design choices associated with transformers (Liu et al., 2022a). Swin V2 is a hierarchical shifted-window transformer designed for stable scaling across model and input size (Liu et al., 2022b). We use the official torchvision ImageNet checkpoint to avoid mistaking an undertrained 27M-parameter model for an architectural result. The task head, an early head, and a fourth mask channel are present in both baseline and Dual-RF.

4 Dual Receptive-Field Routing

4.1 Input and supported-field completion

Let $x \in [0, 1]^{C \times H \times W}$ be an image and $m \in \{0, 1\}^{1 \times H \times W}$ be an observation mask, broadcast across channels. The sensory sheet is $u = m \odot x$. For location i and odd field width s , define the center-excluded neighborhood

$$\mathcal{A}_s(i) = \mathcal{N}_s(i) \setminus \{i\}. \quad (1)$$

The observed support and normalized estimate are

$$d_i^{(s)} = \sum_{j \in \mathcal{A}_s(i)} m_j, \quad (2)$$

$$p_i^{(s)} = \frac{\sum_{j \in \mathcal{A}_s(i)} m_j x_j}{\max(1, d_i^{(s)})}. \quad (3)$$

For an ordered set of surround widths $\mathcal{S}$, choose

$$s_i^* = \min\{s \in \mathcal{S} : d_i^{(s)} > 0\}, \quad p_i = p_i^{(s_i^*)}. \quad (4)$$

The completed sheet is

$$\tilde{x} = m \odot x + (1 - m) \odot p. \quad (5)$$

Equation (5) guarantees that no observed value is overwritten. The local Dual-RF operator uses $\mathcal{S} = (3)$ and leaves an unsupported location at zero. The adaptive-reach extension uses $\mathcal{S} = (3, 5, 9, 17, 31)$. Both implementations use static average-pooling kernels, subtract the center contribution, and select support with tensor masks. Neither has learned parameters or per-pixel Python control flow.

4.2 Shared Swin path and early head

Both variants concatenate the normalized image and binary mask, yielding a four-channel input to Swin V2-Tiny. The checkpoint’s RGB stem weights initialize the first three channels; the mask channel initializes to zero. Stages 1–3 produce a feature map

$$h = F_{1:3}([\text{Norm}(v), m]), \quad (6)$$

where $v = u$ for masked Swin and $v = \tilde{x}$ for Dual-RF. A LayerNorm and linear early head produce logits ℓ^e and probabilities $q = \text{softmax}(\ell^e)$.

4.3 Fixed-capacity final-stage route

At inference, the default score is early-head entropy,

$$a_i = H(q_i) = - \sum_c q_{ic} \log(q_{ic} + \epsilon). \quad (7)$$

For batch size B and declared capacity ρ , the active set is

$$I = \text{TopK}(a, \lceil \rho B \rceil). \quad (8)$$

Only the gathered tensor $h[I]$ executes patch merging, stage 4, normalization, and the deep head. Outputs are

$$\ell_i = \begin{cases} F_4(h_i), & i \in I, \\ \ell_i^e, & i \notin I. \end{cases} \quad (9)$$

This is one dense operation on a fixed-size subbatch. Identities vary, but tensor size does not. We also replace entropy with deterministic random scores and with an oracle score that uses early classification error. Oracle results are diagnostic and cannot be deployed.

4.4 Training objective

Training always executes the dense deep path so both heads receive supervision. For labels y , the loss is

$$\mathcal{L} = \text{CE}(\ell^d, y) + \text{CE}(\ell^e, y) + 0.2 D_{\text{KL}}(q_{\text{stop}}^d \| q^e). \quad (10)$$

No reconstruction objective, extra label, or learned completion network is used. This matters for attribution: the only input-representation difference is Equation (5).

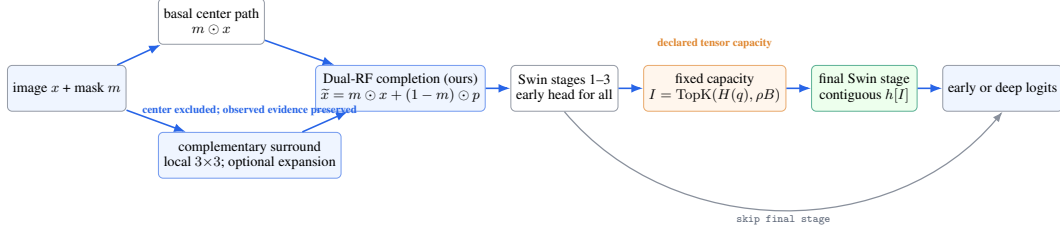


Figure 1: Dual Receptive-Field Routing (Dual-RF, ours). Representation and execution are distinct. The deterministic complementary surround fills only unobserved locations with available context; the adaptive variant expands its field only when local support is absent. Every example executes the shared Swin path, while a fixed-size gathered subbatch executes the final stage.

5 Experimental protocol

5.1 Paired pretrained-Swin studies

The paired studies use natural-color CIFAR-10 and grayscale MNIST at 32×32 , with the official torchvision Swin V2-Tiny ImageNet checkpoint. For each dataset and seed, we sample 6,000 examples from its public training split and 4,000 from its public test split. No augmentation or example selection based on outcomes is used. Baseline and adaptive Dual-RF use the same four-channel patch stem, early head after stage 3, original stage 4, and 10-class heads; each has 27,596,414 learned parameters. The body learning rate is 7.5×10^{-5} , the head rate is 3×10^{-3} , AdamW weight decay is 10^{-4} , and training lasts four epochs with batch size 64.

Training masks cycle among rows, one contiguous square block, and independent pixels. Missingness follows a deterministic schedule from 15% to 25%. Evaluation conditions are: clean input; 45% rows missing; one square block covering approximately 35% of pixels; and 45% pixels missing independently. Evaluation masks are fixed within each paired seed and are not used for hyperparameter tuning. The severe missingness levels are outside the training range.

The paired seeds are 43, 47, and 53. For each seed, model construction is reseeded before each variant; the parameterized modules are instantiated in the same order; data permutations and training masks are identical. On CIFAR-10 we additionally train a direct local-field control with $\mathcal{S} = (3)$ under the same initialization and protocol. It differs from the adaptive variant only in leaving locations without local support unfilled. We report per-seed values, means, sample standard deviations, and 95% Student- t intervals. With $n = 3$, $t_{0.975,2} = 4.303$, so intervals are intentionally wide.

5.2 Latency protocol

Latency is measured end to end on Apple M4 MPS with a 128-example batch drawn from the held-out set under 45% row missingness. Each model receives warm-up passes. Timed measurements run 16 repetitions and synchronize MPS before and after the region. The measurement includes completion, the explicit mask channel, early-head scoring, top- k , gather, final-stage execution, and scatter. It excludes data loading. We report ms/example and paired fractional speedup. This is software latency on one device, not joules or a universal hardware estimate.

5.3 Capacity and routing controls

MNIST Dual-RF is evaluated at $\rho \in \{0.25, 0.50, 0.75, 1.00\}$. Entropy is the deployable route score; random and label-using oracle scores are evaluated at $\rho = 0.25$. CIFAR-10 adaptive and local variants are evaluated at $\rho \in \{0.25, 1.00\}$. If entropy does not beat random, the result supports fixed capacity but not intelligent allocation. If oracle is better, routing headroom exists even when the deployable score is weak.

Table 1: CIFAR-10 Swin V2-Tiny result at 25% final-stage capacity. Values are mean $\pm$ sample sd in percent across three paired seeds. The masked baseline executes its final stage densely. Both completed variants are parameter-identical to it.

Condition	Masked Swin	Local 3×3	Adaptive Dual-RF (ours)	Adaptive-base	Adaptive-local
Clean	76.775 $\pm$ 1.609	79.750 $\pm$ 1.177	79.050 $\pm$ 0.950	+2.275	-0.700
45% rows missing	58.850 $\pm$ 0.553	68.958 $\pm$ 1.125	71.175 $\pm$ 2.580	+12.325	+2.217
35% block missing	64.367 $\pm$ 0.929	66.633 $\pm$ 0.732	65.908 $\pm$ 1.482	+1.542	-0.725
45% pixels missing	45.867 $\pm$ 3.333	72.933 $\pm$ 0.345	72.908 $\pm$ 0.988	+27.042	-0.025

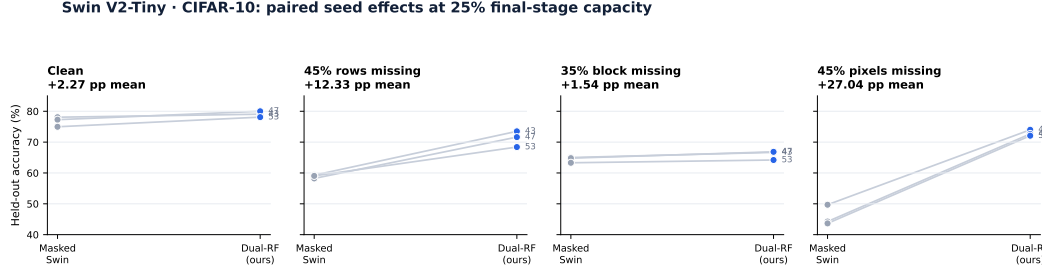


Figure 2: CIFAR-10 paired per-seed effects for adaptive Dual-RF (ours) at 25% final-stage capacity. A shared vertical scale across all four panels prevents small changes from appearing comparable to the row and pixel effects. Each line is one paired seed.

5.4 ConvNeXt-style replication and historical controls

A smaller paired ConvNeXt-style body uses 12,000 training and 4,000 held-out MNIST examples, 12 epochs, three seeds, and the same four evaluation conditions. Its Dual-RF representation uses a single 5×5 center-excluded field; its final stage has six blocks and is routed at 50% capacity. It adds a 194-parameter value router, a 0.020% increase, so it is near-matched rather than exact. This replication predates the final Swin simplification and is reported to show where effects shrink.

Historical PBN temporal, older ConvNeXt, and MLP-gated Swin pilots are moved to the appendix. They use different mechanisms and protocols and are not pooled with Dual-RF.

6 Results

6.1 Natural-color CIFAR-10: sizeable robustness at fixed capacity

Table 1 gives the principal natural-image result. Relative to dense masked Swin, adaptive Dual-RF at 25% final-stage capacity changes clean, row, block, and pixel accuracy by +2.275, +12.325, +1.542, and +27.042 percentage points. Every seed improves in all four conditions. The paired 95% intervals are $[-0.566, +5.116]$, $[+5.843, +18.807]$, $[+0.085, +2.999]$, and $[+21.143, +32.940]$ points, respectively. Rows, blocks, and pixels exclude zero, but three runs remain a small confirmatory sample.

The single- 3×3 control changes the causal interpretation. It reaches 79.750, 68.958, 66.633, and 72.933 percent, recovering nearly all absolute improvement. Adaptive-minus-local means are -0.700, +2.217, -0.725, and -0.025 points. Their paired intervals all include zero; even the positive row mean has interval $[-2.940, +7.374]$. The supported mechanism is local evidence-preserving completion. Conditional expansion remains a hypothesis, not an established gain.

6.2 MNIST transfer: repeated row/pixel gains and a block boundary

Table 2 reports the independent MNIST result at 25% final-stage capacity. Adaptive Dual-RF improves every seed under missing rows: +2.85, +8.40, and +3.23 points. It also improves every seed under missing pixels: +3.50, +6.70, and +3.00 points. Clean gains are +0.73, +0.80, and +0.03 points. Large-block gains are +0.23, +0.68, and -2.15 points, yielding a negative mean. Unlike CIFAR-10, a broad contextual average does not reliably preserve digit topology inside a large block.

Table 2: MNIST paired Swin V2-Tiny result at 25% final-stage capacity. Values are mean $\pm$ sample sd in percent across three seeds. “Gain” is the paired adaptive Dual-RF minus masked-Swin mean in percentage points.

Condition	Masked Swin	Dual-RF (ours)	Gain (pp)
Clean	97.925 $\pm$ 0.254	98.442 $\pm$ 0.263	+0.517
45% rows missing	89.158 $\pm$ 2.654	93.983 $\pm$ 0.869	+4.825
35% block missing	70.167 $\pm$ 3.060	69.750 $\pm$ 1.583	−0.417
45% pixels missing	93.325 $\pm$ 1.905	97.725 $\pm$ 0.265	+4.400

Table 3: Dual-RF capacity sweep. Accuracy is the mean across seeds; severe accuracy averages row, block, and pixel conditions.

Final stage	Latency	Clean	Rows	Block	Pixels
25%	0.946	98.442	93.983	69.750	97.725
50%	1.087	98.442	93.983	69.725	97.725
75%	1.225	98.442	93.983	69.725	97.725
100%	1.369	98.442	93.983	69.725	97.725
Masked dense	1.365	97.925	89.158	70.167	93.325

The 95% paired-gain intervals are $[-0.55, 1.58]$, $[-2.88, 12.53]$, $[-4.19, 3.35]$, and $[-0.59, 9.39]$ points for clean, rows, block, and pixels. All include zero because three seeds give only two degrees of freedom. We therefore describe the repeated direction and magnitude, not statistical significance. The reduced standard deviation of Dual-RF under row and pixel masks is encouraging but also requires larger replication.

6.3 Fixed capacity produces a stable latency reduction

On CIFAR-10, dense masked Swin takes 1.375 ms/example and adaptive Dual-RF at 25% takes 0.974 ms/example. The paired speedup is 29.2% with interval $[28.70, 29.61]\%$. The local- 3×3 control takes 0.942 ms/example and is 31.5% faster than baseline. Thus the simpler, better-supported representation is also the faster completed variant in this implementation.

On MNIST, masked Swin takes 1.365 ± 0.017 ms/example and Dual-RF takes 0.946 ± 0.014 ms/example. Paired speedups are 30.56%, 30.90%, and 30.66%, giving a mean of 30.7% and a 95% interval of $[30.27, 31.13]\%$. Dense Dual-RF takes 1.369 ms/example, nearly the same as the 1.365 ms masked baseline; the complementary field’s pooling overhead is small relative to Swin.

The flat MNIST accuracy across 25–100% capacity is scientifically important. The final Swin stage rarely changes the predicted class on this transfer task. The speed result is measured, but it may reflect task simplicity rather than a universally superior route. An ImageNet-scale study could require much more final-stage capacity.

6.4 Entropy routing is close to random

At 25% capacity, entropy minus random accuracy is -0.008 , $+0.125$, $+0.150$, and $+0.142$ points for clean, rows, block, and pixels. These differences are negligible relative to seed variability. Oracle routing adds little for clean, rows, and pixels because early and deep predictions agree, but it improves block accuracy from 69.75% to 72.99%. Thus, large blocks expose routing headroom; entropy does not identify it reliably.

This control changes the method claim. Dual-RF does not currently demonstrate a brain-derived allocation signal that beats a generic router. The robust contribution is the complementary representation and fixed execution capacity. The website labels the router hypothesis “not supported.”

6.5 Smaller ConvNeXt replication

The ConvNeXt-style three-seed experiment produces paired dense Dual-RF gains of $+0.07$, $+0.67$, $+0.58$, and $+0.58$ points for clean, rows, block, and pixels. At 50% final-stage capacity, it is 15.6%

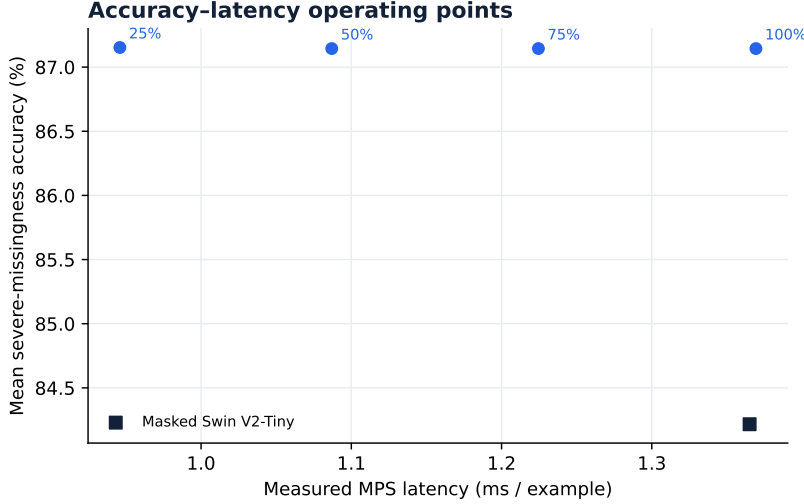


Figure 3: Measured accuracy–latency operating points. The 25% route dominates denser Dual-RF settings in this protocol, but this does not establish the same frontier for larger datasets.

faster than the dense masked baseline. The mismatch-aware router differs from a learned confidence-only router by 0.00, 0.00, +0.06, and 0.00 points across the four conditions. This independently rejects the claim that observed center-excluded mismatch is adding useful routing information in the current implementation.

The smaller effects also caution against treating the Swin magnitude as a property of the operator alone. Pretraining, backbone hierarchy, early-head quality, and optimization interact with completion. The two studies support a framework, not a universal effect size.

7 Analysis and ablations

7.1 Mask topology and dataset interact

For row and independent-pixel masks, most missing locations have nearby observed support. The local operator therefore interpolates from adjacent evidence while suppressing zero-mask discontinuities. This explains why the 3×3 control already captures the large CIFAR-10 gains. It also explains why adaptive reach has only a positive mean on rows: a fully missing row sometimes has no immediate vertical support at one side of an image, but nearby rows are usually sufficient.

Large blocks expose the limit of spatial averaging. The local operator fills only a thin supported boundary and leaves the unsupported interior zero. The adaptive ladder instead inserts averages from 17×17 or 31×31 fields. On CIFAR-10 both completed variants improve over baseline, but local completion has the higher mean; on MNIST the adaptive mean falls below baseline. Broad reach therefore does not behave like semantic inpainting, and its effect is dataset-dependent.

7.2 Causal component ledger

7.3 Why exact parameter matching matters

The completion operator has no trainable weights. The masked baseline nevertheless contains the same early head and four-channel stem as Dual-RF, so both models have 27,596,414 parameters. This removes three common confounds: added capacity, a different classifier, and a different checkpoint function. Runtime differs only because Dual-RF adds static pooling and routes the final stage.

The ConvNeXt replication is labeled near-matched because its tiny value router adds 194 parameters. The old PRM comparison described a 1.38% larger model as “parameter-matched” in prose; that wording was incorrect and has been removed.

Table 4: Claim ledger. “Supported” means supported by the current paired protocol, not established generally.

Claim	Direct test	Status
Local complementary completion helps	Exact-parameter CIFAR-10 single-field control	Supported under all tested CIFAR masks
Adaptive reach beats one local field	Adaptive ladder vs exact single- 3×3 control	Not supported generally
Fixed capacity maps to latency	End-to-end gather–compute–scatter timing	Supported on M4 MPS
Prediction mismatch improves routing	Mismatch-aware vs confidence-only router	Not supported
Entropy allocates depth intelligently	Entropy vs deterministic random route	Not supported clearly
Broad averages repair missing interiors	Adaptive vs local block control	Not supported
Energy is reduced	Direct joule measurement	Not tested
Broad vision SOTA improves	ImageNet-scale benchmark	Not tested

7.4 Software activity, latency, and energy

Selective computation has at least three accounting levels. Activity records how many examples or tokens enter a branch. Latency measures one software/runtime/device combination. Energy measures joules under a controlled power protocol. Dual-RF reports the first two. The 25% capacity is not itself a 75% model-level FLOP reduction because stages 1–3 still execute densely. The 30.7% latency reduction is stronger evidence than an activity proxy, but it does not imply the same reduction in energy, cloud cost, or another accelerator.

8 Limitations and broader impacts

Scale. The strongest study uses a 6,000-example CIFAR-10 training subset and 4,000 held-out images; MNIST uses the same sizes. The official pretrained Swin backbone is strong, but this is not full-data CIFAR-10, CIFAR-100, or ImageNet state of the art. Small transfer tasks can make a late stage unusually redundant, inflating the safe speedup from early exit.

Synthetic missingness. Masks are controlled and explicit. Real sensor failures can be correlated with content, unknown to the model, or replaced by adversarial artifacts. Dual-RF does not infer m and should not be presented as generic corruption robustness.

Statistical power. Three paired seeds reveal direction and variability but yield wide accuracy intervals. A publication-grade confirmatory study should preregister at least ten seeds or use a substantially larger test set with hierarchical uncertainty analysis.

Prior work and novelty. Masked normalized convolution, early exits, and fixed top- k capacity all have direct precedents. Surprise-based fixed-capacity transformers are especially close. The direct 3×3 control further narrows the claim: adaptive reach is not clearly superior to local completion. The contribution is an evidence-preserving center/surround formulation, exact integration into a strong backbone, and a paired accuracy–latency audit. A reviewer may reasonably judge that combination incremental without full-scale natural-image and partial-convolution comparisons.

Biological scope. The method is not a neuron model. It does not reproduce dendrites, spikes, laminar feedback, synaptic plasticity, or metabolic allocation. The biological claim is limited to a computational correspondence between complementary spatial fields.

Deployment. A fixed route can silently use an early prediction on difficult inputs. Production use should log capacity, route score, head disagreement, and downstream confidence, and should provide a dense fallback. In safety-critical vision, a latency optimization is not sufficient evidence for deployment.

Broader impact. Hardware-aligned selective depth could reduce inference latency for edge systems and long-running visual services. It could also make behavior harder to audit if branch decisions are hidden. Publishing negative route controls and per-seed values is part of mitigating that risk.

9 Conclusion

Dual-RF begins with one measured visual-cortex motif: feedback can define a contextual receptive field complementary to feedforward drive. Its engineering translation preserves observed evidence, fills missing locations only from observed surround context, and executes late depth at a fixed tensor capacity. On paired CIFAR-10 subsets, adaptive Dual-RF improves row and pixel missingness by +12.325 and +27.042 points while reducing measured latency by 29.2%; MNIST independently repeats the row/pixel direction. The decisive control is simpler: one local 3×3 surround recovers nearly all CIFAR improvement and is 31.5% faster. Adaptive reach and entropy routing are not established. The next decisive test is full-scale natural imagery with partial-convolution controls, naturally missing evidence, larger replication, compiled routes, and direct power measurement.

A Per-seed primary results

Table 5: All CIFAR-10 paired accuracies (percent). Baseline is dense; both Dual-RF variants use 25% final-stage capacity. Nothing is selected by best epoch or seed.

Seed	Model	Clean	Rows	Block	Pixels
43	Masked Swin	78.075	59.275	65.000	49.700
43	Local Dual-RF (ours)	79.350	70.225	67.475	73.325
43	Adaptive Dual-RF (ours)	79.050	73.500	66.675	74.000
47	Masked Swin	77.275	58.225	64.800	44.250
47	Local Dual-RF (ours)	81.075	68.075	66.150	72.800
47	Adaptive Dual-RF (ours)	80.000	71.625	66.850	72.650
53	Masked Swin	74.975	59.050	63.300	43.650
53	Local Dual-RF (ours)	78.825	68.575	66.275	72.675
53	Adaptive Dual-RF (ours)	78.100	68.400	64.200	72.075

Table 6: CIFAR-10 per-seed MPS latency. Each completed variant executes 25% of examples through the final stage.

Seed	Masked dense	Adaptive	Local 3×3	Adaptive speedup	Local speedup
43	1.4044	0.9976	0.9462	28.97%	32.63%
47	1.3619	0.9646	0.9556	29.18%	29.83%
53	1.3597	0.9609	0.9248	29.33%	31.98%

Table 7: All MNIST paired accuracies (percent). Nothing is selected by best epoch or best seed.

Seed	Model	Clean	Rows	Block	Pixels
43	Masked Swin	97.975	91.450	68.350	94.425
43	Dual-RF (ours), 25%	98.700	94.300	68.575	97.925
47	Masked Swin	97.650	86.250	68.450	91.125
47	Dual-RF (ours), 25%	98.450	94.650	69.125	97.825
53	Masked Swin	98.150	89.775	73.700	94.425
53	Dual-RF (ours), 25%	98.175	93.000	71.550	97.425

Table 8: MNIST per-seed latency and speedup at 25% final-stage capacity.

Seed	Masked dense (ms/ex)	Dual-RF 25% (ms/ex)	Speedup
43	1.3852	0.9619	30.56%
47	1.3555	0.9367	30.90%
53	1.3546	0.9393	30.66%

B Router controls

Table 9: Routing controls at 25% capacity, mean accuracy in percent. Oracle uses labels and is not deployable.

Condition	Entropy	Random	Oracle
Clean	98.442	98.450	98.442
Rows	93.983	93.858	93.983
Block	69.750	69.600	72.992
Pixels	97.725	97.583	97.725

The oracle’s block advantage means the final stage can repair some early-head mistakes, but entropy does not rank those examples correctly. A future router should be evaluated against random at equal

capacity and should report the paired probability that it selects an example with positive deep-minus-early utility.

C Reference pseudocode and invariants

```

Input:  $x, m, \mathcal{S}, \rho$  with  $\mathcal{S} = (3)$  or  $(3, 5, 9, 17, 31)$ 
 $u \leftarrow m \odot x$ 
 $p \leftarrow 0, \quad r \leftarrow 1 - m$ 
for  $s \in \mathcal{S}$  :
     $n_s \leftarrow \text{BoxSum}_s(u) - u$ 
     $d_s \leftarrow \text{BoxSum}_s(m) - m$ 
     $v_s \leftarrow \mathbf{1}[d_s > 0], \quad z_s \leftarrow r \odot v_s$ 
     $p \leftarrow p + z_s \odot n_s / \max(1, d_s)$ 
     $r \leftarrow r \odot (1 - v_s)$ 
 $\tilde{x} \leftarrow m \odot x + (1 - m) \odot p$ 
 $h, \ell^e \leftarrow F_{1:3}([\text{Norm}(\tilde{x}), m])$ 
 $I \leftarrow \text{TopK}(H(\text{softmax}(\ell^e)), \lceil \rho B \rceil)$ 
 $\ell \leftarrow \ell^e; \quad \ell[I] \leftarrow F_4(h[I])$ 
return  $\ell, I$ 

```

Figure 4: Backbone-independent Dual-RF (ours) inference. The local variant uses one field; the adaptive variant uses the ladder. Box sums compile to dense pooling/convolution.

The implementation is unit-tested conceptually by the following invariants:

1. $m_i = 1 \Rightarrow \tilde{x}_i = x_i$ exactly.
2. The center coefficient is zero at every scale.
3. A location receives at most one field estimate, from the first supported scale.
4. $|I| = \lceil \rho B \rceil$ for $0 < \rho < 1$.
5. At $\rho = 1$, routed logits equal dense deep logits.
6. The mask-channel stem initializes to zero, preserving the RGB checkpoint function before fine-tuning.

D Audit trail from the predecessor method

The predecessor report centered Predictive Retinotopic Memory (PRM). Its full adapter used a $2 \rightarrow 16 \rightarrow 16 \rightarrow 1$ predictor with two passes. The model called “hardware PRM” instead used a $2 \rightarrow 8 \rightarrow 1$ predictor and a separate 12-channel refiner. It was evaluated for one seed. Therefore, the reported 21% speed difference between full and hardware variants jointly changed width, depth, parameter count, and routing. It was not causal evidence for routing.

The predecessor mismatch score also had a training inconsistency. The reconstruction loss was applied after observed values were clamped, so observed pixels provided no direct reconstruction gradient to calibrate the predictor used for observed mismatch. The current Swin method removes that learned mismatch score and completion loss entirely. The ConvNeXt audit explicitly compares mismatch-aware and confidence-only routers and finds no material advantage.

The predecessor paper used the phrase “parameter-matched” for a model with 1.38% additional parameters. The present report uses “exact” only for the Swin comparison and “near-matched” for the 0.020% ConvNeXt difference.

E Historical experiments retained as boundaries

Earlier work used a different Predictive-Budget Neuron (PBN) state packet for temporal streams. On a five-seed public MNIST row stream, PBN achieved 91.765% clean and 85.445% stress accuracy versus 90.295% and 83.725% for a GRU, with fewer measured parameters. On a synthetic event-order stream, the PBN gain was much larger, but task construction favored predictive state. These results motivate missing-evidence studies but do not validate Dual-RF.

An older pretrained Swin V2-Tiny intervention split each MLP into two branches and used a local feature residual gate. Across three seeds it changed clean accuracy from 97.625% to 97.350%, stress accuracy from 74.042% to 73.967%, and latency from 1.382 to 1.662 ms/example. This negative result is retained because it demonstrates that inserting a “brain-inspired” gate into a strong backbone can make both accuracy and latency worse. Dual-RF differs by intervening at explicit missing evidence and by routing a whole final stage as a fixed-size tensor.

F Reproducibility and artifact map

The project page is hosted at <https://alex-dils.com/brain-inspired-ai/>. The following local artifacts generate the report:

- `swin_dual_rf_routing_experiment.py`: paired Swin training, stress evaluation, routing controls, and latency.
- `swin_single_scale_control.py`: parameter-identical local- 3×3 CIFAR-10 control.
- `apical_predictive_routing_experiment.py`: smaller ConvNeXt audit.
- `swin_drf_cifar10_seed{43,47,53}.json`: CIFAR-10 baseline and adaptive per-seed observations.
- `swin_single3_cifar10_seed{43,47,53}.json`: direct local-field control observations.
- `swin_drf_mnist_seed{43,47,53}.json`: full per-seed Swin observations.
- `drfr_mnist_seed{43,47,53}.json`: full per-seed ConvNeXt observations.
- `build_dual_rf_artifacts.py`: statistics, vector figures, website payload, and generated LaTeX macros.
- `dual-rf-results.json`: website payload preserving every plotted seed.

The public page labels the method “Dual-RF (ours),” identifies the fMRI and calcium-imaging sources, and exposes the adaptive-field, block, and routing boundaries.

G Recommended confirmatory study

A stronger submission should preregister the following factorial design:

1. datasets: full CIFAR-100, ImageNet-1k, and a naturally missing sensor dataset;
2. completion: none, single-scale partial convolution, Dual-RF ladder, learned inpainting at matched FLOPs;
3. route: random, entropy, learned utility, and oracle diagnostic;
4. capacity: 0, 25, 50, 75, and 100%;
5. masks: known row/pixel/block, unknown corruption, and content-dependent sensor dropout;
6. systems: eager, compiled, and fused implementations with latency, peak memory, bytes moved, and joules;
7. statistics: at least ten seeds for small data or a hierarchical bootstrap over examples and training runs.

The hypothesis should be weakened if a single-scale partial convolution matches Dual-RF, if random routing matches a learned router on natural data, if the speedup disappears after compilation of the dense baseline, or if gains require a mask unavailable at deployment.

H NeurIPS-style checklist

1. **Claims:** All headline claims state the dataset, route capacity, and measured metric. Negative block/router results are included.
2. **Limitations:** Dataset scale, synthetic masks, statistical power, related prior art, and lack of energy measurement are discussed.
3. **Theory:** No theorem or formal generalization guarantee is claimed.
4. **Reproducibility:** Equations, hyperparameters, seeds, device, batch shape, pseudocode, per-seed values, and code artifact names are provided.
5. **Data:** CIFAR-10, MNIST, and pretrained torchvision weights are public. Real biological visuals are separately sourced and not used for model fitting.
6. **Compute:** Each Swin study trains exact 27.6M-parameter variants for four epochs on 6,000 examples per seed; CIFAR-10 includes an additional single-field control. Latency is measured on Apple M4 MPS.
7. **Human subjects:** No new human-subject data are collected. The displayed fMRI comes from an existing open dataset and is not analyzed for a model claim.
8. **Code of ethics:** The report avoids equating biological images with architecture outputs and avoids interpreting latency as energy.

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